Satellite navigation

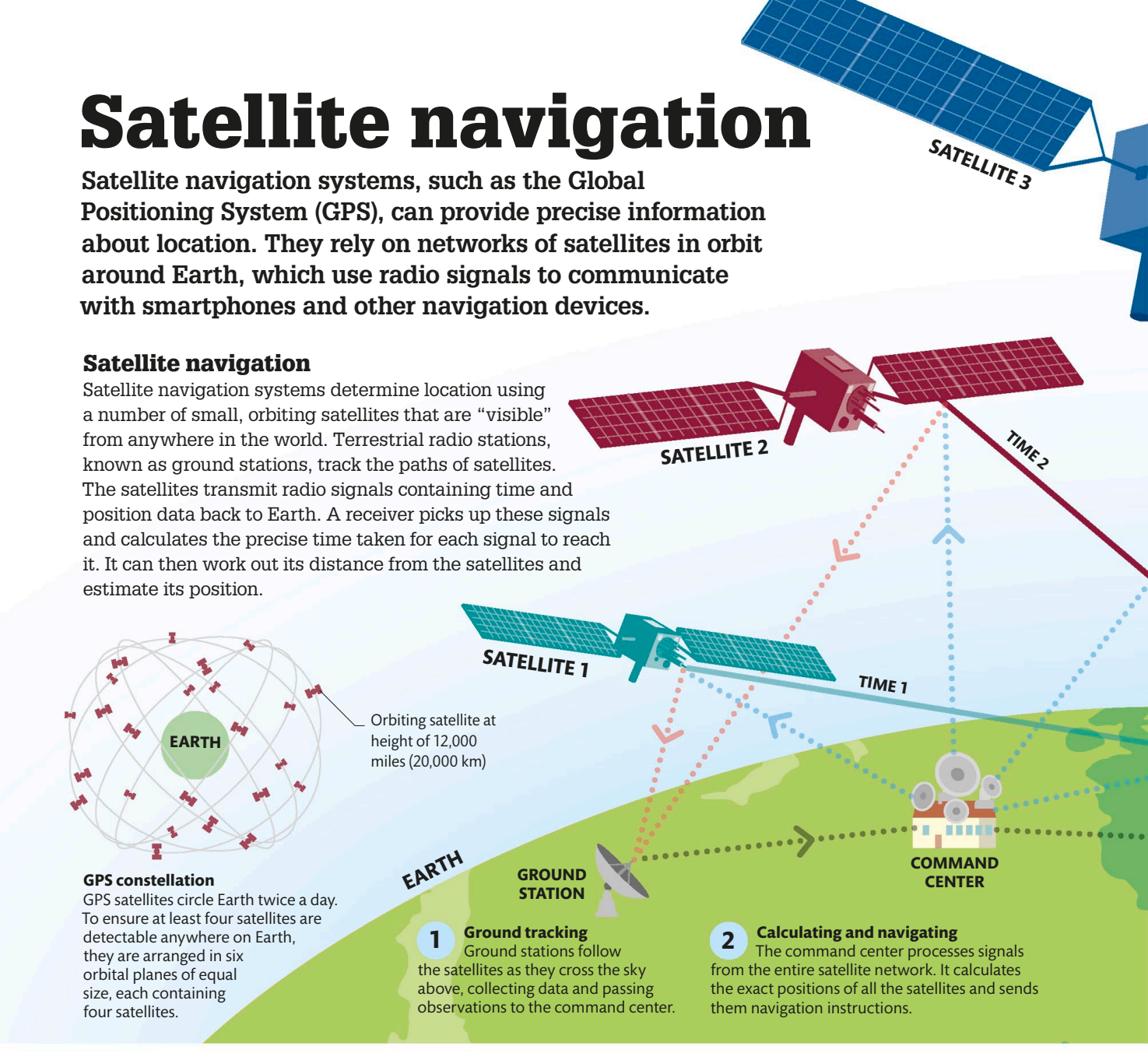
Satellite navigation systems, such as the Global Positioning System (GPS), can provide precise information about location. They rely on networks of satellites in orbit around Earth, which use radio signals to communicate with smartphones and other navigation devices.

Satellite navigation

Satellite navigation systems determine location using a number of small, orbiting satellites that are "visible" from anywhere in the world. Terrestrial radio stations, known as ground stations, track the paths of satellites. The satellites transmit radio signals containing time and position data back to Earth. A receiver picks up these signals and calculates the precise time taken for each signal to reach it. It can then work out its distance from the satellites and estimate its position.

Orbiting satellite at height of 12,000 miles (20,000 km)

GPS constellation
GPS satellites circle Earth twice a day. To ensure at least four satellites are detectable anywhere on Earth, they are arranged in six orbital planes of equal size, each containing four satellites.



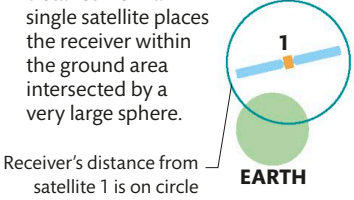
1 Ground tracking
Ground stations follow the satellites as they cross the sky above, collecting data and passing observations to the command center.

2 Calculating and navigating
The command center processes signals from the entire satellite network. It calculates the exact positions of all the satellites and sends them navigation instructions.

How trilateration works

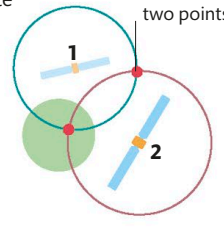
Calculating the distance from one satellite places the receiver somewhere within a sphere around it. Finding the distance from other satellites narrows its possible position down to the area where the spheres intersect. This process is called trilateration.

Satellite 1
Calculating the distance from a single satellite places the receiver within the ground area intersected by a very large sphere.



Receiver's distance from satellite 1 is on circle

Satellite 2
Finding its distance from a second satellite reduces the possible area in which the receiver may be positioned to two points on an intersecting line.



Location narrowed down to either of two points